United International University
B.Sc. in Computer Science & Engineering (CSE)
CSE 2215: Data Structure and Algorithms-I

Term Final Exam: Spring 2026   Time: 2 Hours   Marks: 40

Answer all questions. Marks are indicated in the right side of each question.
[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

Q1) **a)** In the FIFA World Cup match, fans enter the stadium through a single turnstile. They line up outside and the turnstile admits them strictly in arrival order: the first fan to reach the line is the first one let in. Identify the ordering principle used in this entry line and name the corresponding data structure. **[1]**

b) The stadium system logs each admitted fan's seat-block number into an array A, in exactly the order they enter, giving **A = [1, 5, 6, 2].** For a post-match security review, the staff needs the list in reverse entry order. Name another data structure that achieves this. Simulate how that structure reverses the array **A** by showing each step. **[1+3]**

Q2) **a)** Consider the following infix expression and answer the questions below:

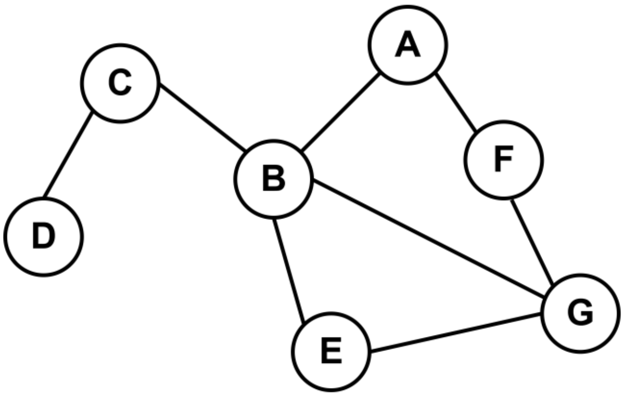
$$(((\boldsymbol{A} + \boldsymbol{B}) * \boldsymbol{C}) \,/\, (\boldsymbol{D} - \boldsymbol{E}))\ \hat{}\ \boldsymbol{F}$$

i. Convert the infix expression to its equivalent postfix expression. **[2]**

ii. Evaluate the obtained postfix expression using a stack, where **A = 2, B = 3, C = 4, D = 10, E = 2, and F = 2.** **[2]**

b) A Tower of Hanoi puzzle with **3** disks and **3** pegs requires **7** moves to transfer all disks from the source peg to the destination peg. If a fourth peg is introduced, would the number of moves increase, decrease, or remain the same? Justify your answer. **[1]**

Q3) The graph below models a computer network in which each vertex **(A–F)** represents a computer, and each edge represents a direct connection between two computers. Every edge is undirected. During traversal, nodes are explored in lexicographical order.

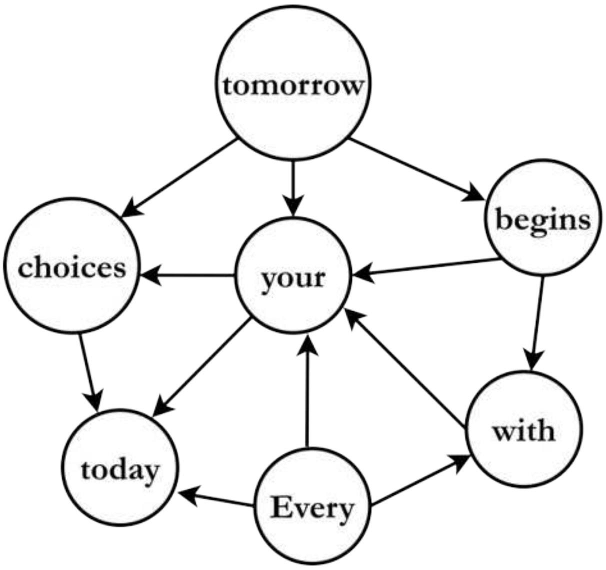


a) A secret message is to be transmitted from A to E as quickly as possible. Assuming that traversing each edge in the network requires the same amount of time, use the suitable algorithm to determine the fastest route from A to E and simulate the algorithm. **[2.5]**

b) When a developer decided to model this graph, he chose to use Adjacency List. Later a junior developer changed it to Adjacency Matrix. Does it impact the time complexity of the Depth First Search Algorithm? Analyze the Time Complexity. **[2]**

c) Perform topological sorting on the following directed graph and provide the final topological ordering. **[3.5]**

[please turn over]



Q4) **a)** Construct a tree satisfying all of the following requirements: **[3]**

- Height = **3**
- Exactly **5** leaf nodes
- Exactly **4** internal nodes
- Root has two children

Draw one possible solution.

b) A tree contains the nodes as follows: **[3]**

10, 20, 30, 40, 50, 60, 70

Arrange the nodes in a binary tree such that:

- Preorder = ascending order
- Postorder = descending order

Draw the tree that satisfies the above requirements.

Q5) You are designing a network router firmware. The router stores active device routing keys in a Binary Search Tree for rapid lookup and sequential diagnostics.

a) Insert the following keys into an initially empty BST in this exact order: **[3]**

45, 12, 85, -53, 30, 62, 86, 35, 83, 98

Draw the final BST formed after all keys are inserted.

b) Insert a new key **-29** in the BST. **[1]**

c) Delete the root key from the BST. **[3]**

d) Explain how you would sort the keys from the BST in ascending order. **[1]**

Q6) Consider the following array representation of a min binary heap where **X** is an unknown value:

H = [6, 10, 8, 15, X, 11, 9, 20, 18, 16]

Important instruction: Each sub-question should be answered independently from the original given heap. Do not use the modified heap from a previous sub-question.

a) Represent the given min heap in complete binary tree form. **[2]**

b) What condition must **X** satisfy so that **H** is a valid min heap? Explain. **[3]**

c) Considering the given heap, if the root **6** is deleted once, can **X** become the new root? Explain. **[3]**